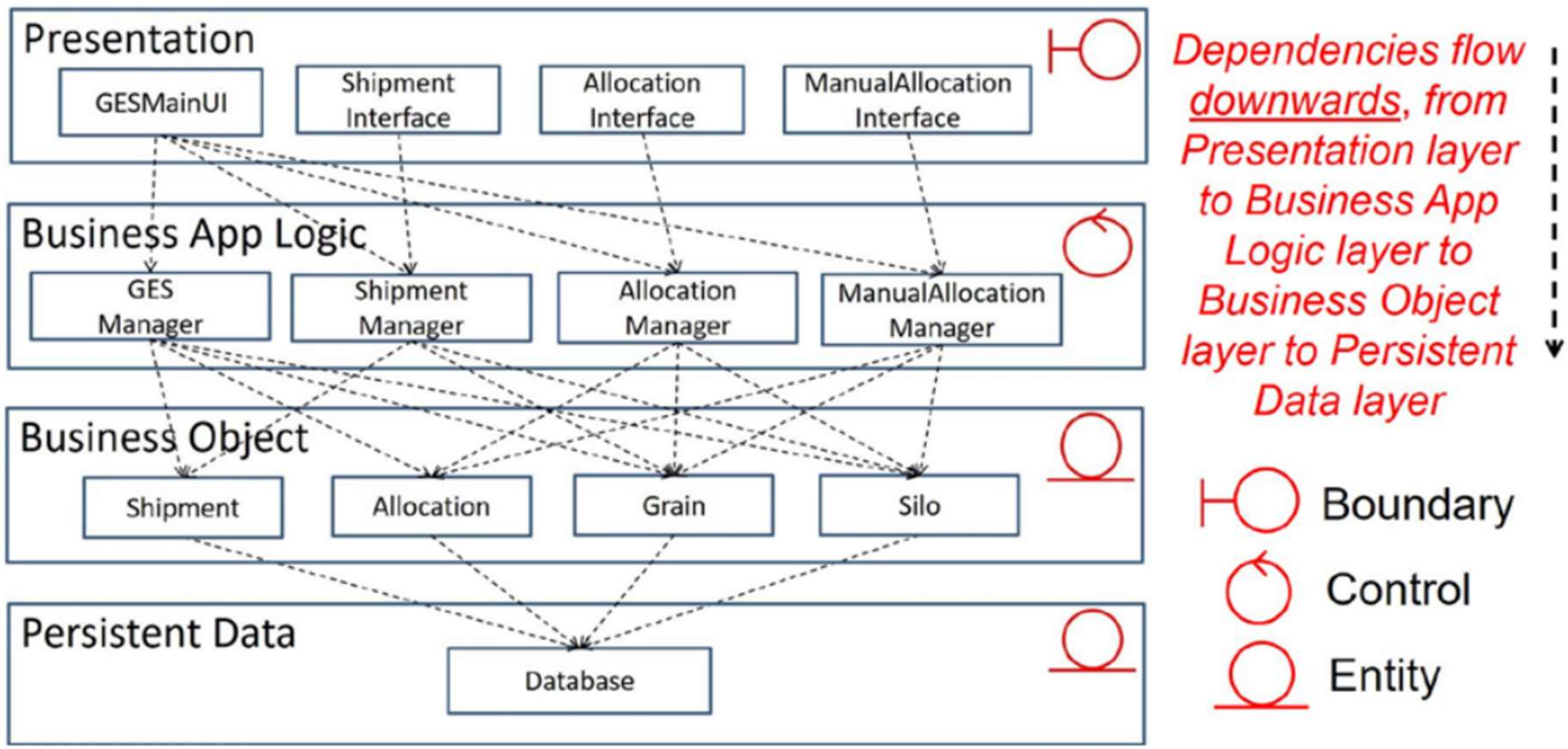
Tutorial 6: Software Architecture and Strategy Pattern

Question 1: Dependencies among boundary, control, and entity objects (cont'd)

(a) Layered Architecture (four-layer architecture style)



Question 2(b):Design problem and solution

What is Strategy Pattern:

- **A strategy is essentially an algorithm (instructions in code) that is implemented to achieve a specific goal.**

- **When you have a family of algorithms, all providing common behavior and you want to change the algorithm without impacting the client code that calls it then you can use strategy pattern.**

- **If there are a number of methodologies (strategies) available to solve a problem then strategy design pattern can be implemented. It enables us to switch implementations for different situations.**

- **It also allows us to add new strategies without recoding and testing all parts of the system.**

Question 2(b):Design problem and solution

Design problem:

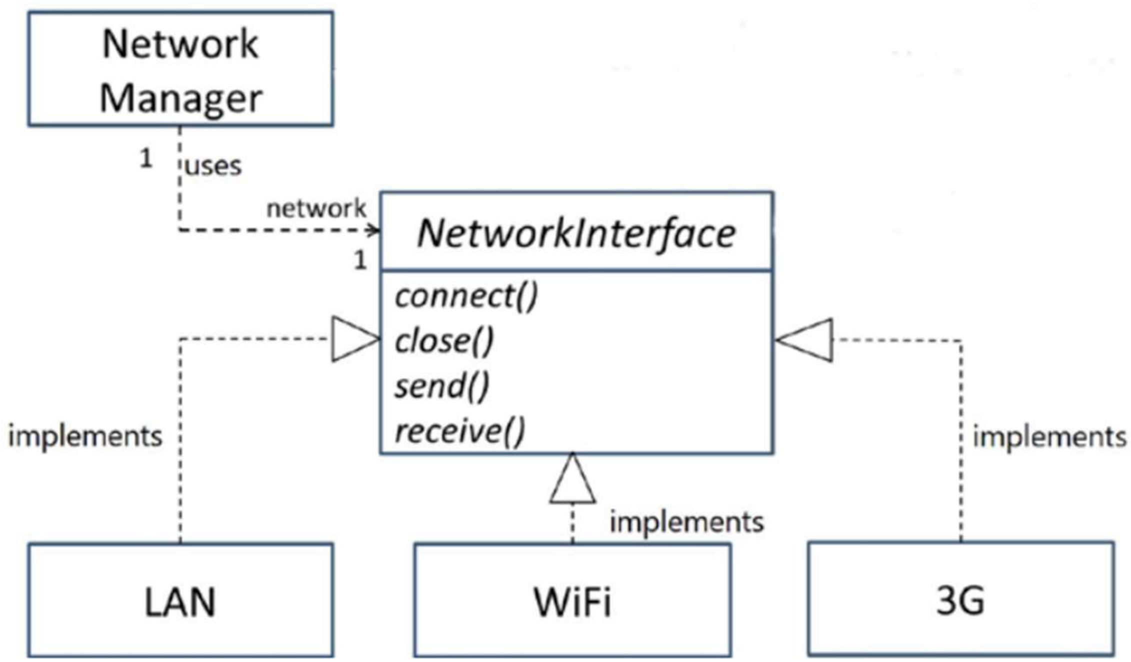
- **Different network protocols should be interchangeable and transparent from the other parts of the mobile device system. Which protocol to use should be easily interchangeable.**
- **Future network protocols should be able to be added with minimal impacts to the application – extensibility.**

Solution: Strategy pattern.

In computer programming, the strategy pattern (also known as the policy pattern) is a behavioral software design pattern that enables selecting an algorithm at runtime. Instead of implementing a single algorithm directly, code receives run-time instructions as to which in a family of algorithms to use.

Strategy lets the algorithm vary independently from clients that use it.

Question 2(c): Class diagram of the strategy pattern



NetworkManager: Context
NetworkInterface: Strategy interface
LAN, WiFi, 3G, etc.: Concrete strategy objects